Both Cycles in One Space

The Spheres of Dharma and their polar reversal sharing one space: the toggle between two exact settings, why they exclude each other, and the interlock, both cycles cut into one solid of convex pieces



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Draft 1 · 16 October 2026 · DICTO DOI: on publication · Licence CC BY 4.0 · Data, scripts, exact checks and renders are included with this record and at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-interlock and studies/2026-10-16-polar-dharma; shared checks in studies/_lib).
The Spheres of Dharma, turning them inside out, the idea of both cycles in one space as a toggle, the question whether the ends could be bent to give both settings, the interlock as one real solid, the fixed viewing bearings and the names of the pieces are DICTO's. Computation, exact arithmetic and renders were done with Claude (Anthropic) as a research assistant.

Abstract. The Spheres of Dharma nest six icosahedral solids, with the Hecatodihedron as an in-between stage; their polar reciprocal is a second exact cycle in which every nesting is reversed (Inverted Shells). We put both cycles in one space. The sphere of reciprocation is then the one free choice, and polarity with the cycle's φ^3 repeat gives mirror spheres every $\varphi^{3/2}$ in radius. At $\rho^2 = \varphi^2/4$ the two cycles share the icosahedron and dodecahedron; at $\rho^2 = 3\varphi^2/4$ they share the icosidodecahedron and triacontahedron; no sphere gives both, because the two settings differ by a factor 3, which is not a power of φ, and the loose ends of the cycle cannot be bent far enough to close the gap. Cutting every piece of one cycle by every piece of the other gives one solid of convex pieces, exact in $\mathbb{Q}(\sqrt{5})$ with remainder 0 in every piece and layer: 536 pieces of 12 kinds in the first setting and 502 pieces of 11 kinds in the second. The setting moves where the two cycles cut each other: the first cuts the outer roofs and leaves the middle columns whole, the second does the opposite.

1. Two cycles

The Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) is an exact, endlessly repeating nesting of six icosahedral solids: icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger, with the DICTO Hecatodihedron (DICTO DOI: 10.5281/zenodo.23360858) as the closed in-between stage of its fourth layer. Taking that stage as a seventh station and reciprocating every station gives the reversed cycle of *Inverted Shells* (DICTO DOI: 10.5281/zenodo.23405029): dodecahedron ⊂ icosidodecahedron ⊂ Ogdoecontahedron ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ^3 larger, the Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688) being the one new solid. DICTO asked whether the two cycles can share one space, and whether together they make one real solid.

2. The toggle

The sphere of reciprocation is the one free choice. Polarity in a sphere of radius ρ sends radius r to ρ^2/r, so in log-radius it is a reflection about $\log \rho$; the cycle's φ^3 repeat is a translation. Together they give mirror spheres every $\varphi^{3/2}$ in radius: the union of the two

cycles reads the same outward and inward, a palindrome in scale, and the polarity swaps the two cycles.

- **Ends setting**, $\rho^2 = \varphi^2/4 = (3 + \sqrt{5})/8$, mirror sphere radius $\varphi/2$: the polarity sends the original icosahedron to the original dodecahedron, so the two cycles share I and D. The reversed turn nests almost entirely inside the original icosahedron.
- **Middle setting**, $\rho^2 = 3\varphi^2/4$, mirror sphere radius $\sqrt{3}\cdot\varphi/2$: they share the icosidodecahedron and the triacontahedron instead. The reversed cycle is then exactly 3 times its ends-setting size.
- **Never both.** Moving from one setting to the other scales ρ^2 by 3, while the cycle only repeats under powers of φ^3; 3 is not a power of φ, so no sphere gives both. Nor can the loose ends be bent to allow it: the original icosahedron can be scaled only within [1/1.232241, 1.894427] (float) before the nest breaks, and sharing both pairs would need a scale of $3/\varphi^3 \approx 0.708$ or 3.

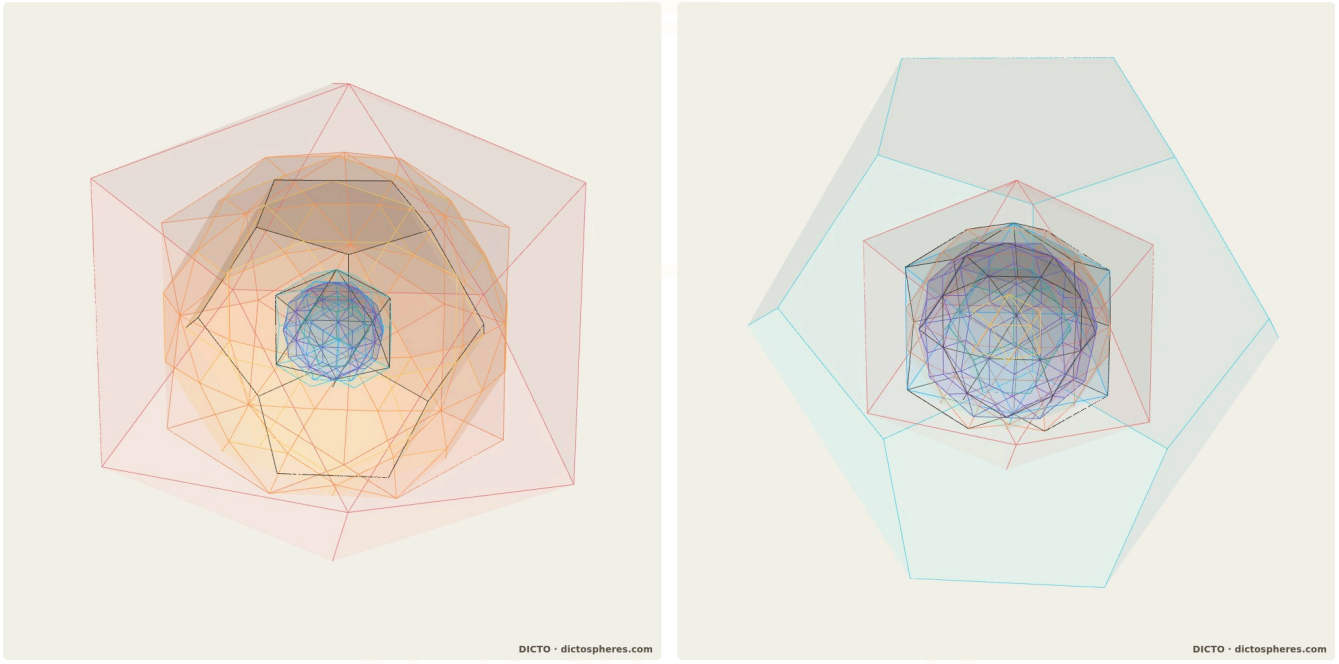


Figure 1. Both cycles in one space, cut in half, see-through: the original cycle warm, the reversed cycle cool, shared stations in dark ink. (a) Ends setting, sharing the icosahedron and dodecahedron; (b) middle setting, sharing the icosidodecahedron and triacontahedron.

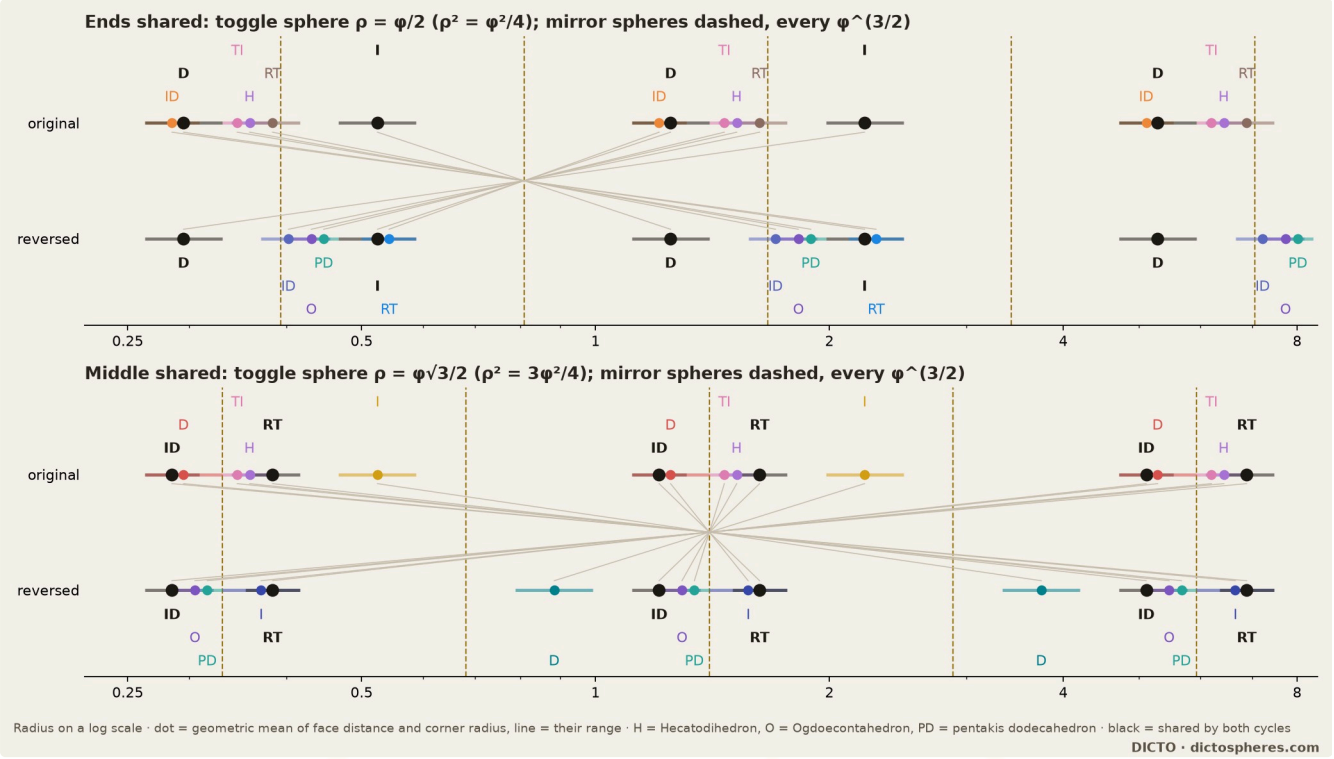


Figure 2. Both settings on one radial scale (log radius, three turns of each cycle). Each station is drawn as the range from its face distance to its corner radius, with a dot at their geometric mean; grey lines join each station to its polar image through the toggle sphere; dashed lines are the mirror spheres, every $\varphi^{3/2}$; black marks the stations the two cycles share. TI = truncated icosahedron, H = Hecatodihedron, O = Ogdoecontahedron, PD = pentakis dodecahedron.

3. The interlock: both cycles as one solid

Each cycle is already cut into convex pieces, by cones from the centre over the faces of each layer's inner solid: 174 pieces in one turn of the original cycle and 154 in one turn of the reversed one. We intersect every piece of one original turn with every piece of the reversed turns that cover it (turns 0 and 1 in the ends setting, −1 and 0 in the middle setting). Every intersection is convex. Every corner was then solved exactly, as the meeting point of three exact planes, after float points lying only on edges were dropped by a rank test on the plane normals. In both settings every original piece is exactly the sum of its parts, and every layer exactly the sum of its interlock pieces, with remainder 0 in $\mathbb{Q}(\sqrt{5})$. Kinds are counted twice, by exact volume and by symmetry orbit of the exact corners, and the two counts agree. DICTO names the two sets as one family from the Inverted Shells project: **DICTO Interlock set 1 (Dharma-Ends)** and **DICTO Interlock set 2 (Dharma-Middle)**. Each piece is named by its set and its number in that set, largest first, followed by its shape: piece I1·1 (Dod) is the largest piece of set 1.

4. Interlock set 1 (Dharma-Ends): 536 pieces, 12 kinds

Several original layers lie wholly inside one reversed layer, so their pieces pass through unchanged; the cutting happens mainly in the outer layer, where the triacontahedron → I·φ^3 roofs are cut by four reversed layers into 240 pieces.

Kind	Count	Original layer	Reversed layer	Corners	Faces (number × sides)	Volume (exact)
I1·1 Dod	12	D → TI	D → ID	15	5 × 3, 7 × 5	-25/24 + 2/3√5
I1·2 TBP	60	RT → I·φ³	PD → I	5	6 × 3	145/1368 + 97/2052√5
I1·3 Hep	60	I → ID	RT → D·φ³	9	3 × 3, 1 × 4, 3 × 5	83/576 − 41/2880√5
I1·4 Hep	60	RT → I·φ³	D → ID	9	2 × 3, 3 × 4, 2 × 5	-11221/5184 + 15629/15552√5
I1·5 Dod	12	H → RT	D → ID	15	5 × 3, 7 × 5	550235/24624 − 245243/24624√5
I1·6 Hep	60	RT → I·φ³	ID → O	9	3 × 3, 1 × 4, 3 × 5	151655/32832 − 602341/295488√5
I1·7 QPy	60	RT → I·φ³	O → PD	5	4 × 3, 1 × 4	41/2592 + 55/7776√5
I1·8 PPy	60	TI → H	D → ID	6	5 × 3, 1 × 5	-11/8 + 5/8√5
I1·9 PPy	12	H → RT	ID → O	6	5 × 3, 1 × 5	-3715/162 + 1663/162√5
I1·10 Tet	60	H → RT	D → ID	4	4 × 3	-93/152 + 21/76√5
I1·11 QPy	60	ID → D	RT → D·φ³	5	4 × 3, 1 × 4	1/576 + 1/576√5
I1·12 Tet	20	I → ID	I → RT	4	4 × 3	-11/48 + 5/48√5

Table 1. Interlock set 1, the ends setting. Layers: I icosahedron, ID icosidodecahedron, D dodecahedron, TI truncated icosahedron, H Hecatodihedron, RT rhombic triacontahedron, O Ogdoecontahedron, PD pentakis dodecahedron; φ³ the next turn out. Corners and faces from the cleaned float corners; volumes exact. Pieces numbered largest first; shapes: Dod 12 faces, Hen 11 faces, Hep 7 faces, Hx 6 faces, TBP triangular bipyramid, HPy hexagonal pyramid, PPy pentagonal pyramid, QPy four-sided pyramid, Tet tetrahedron.

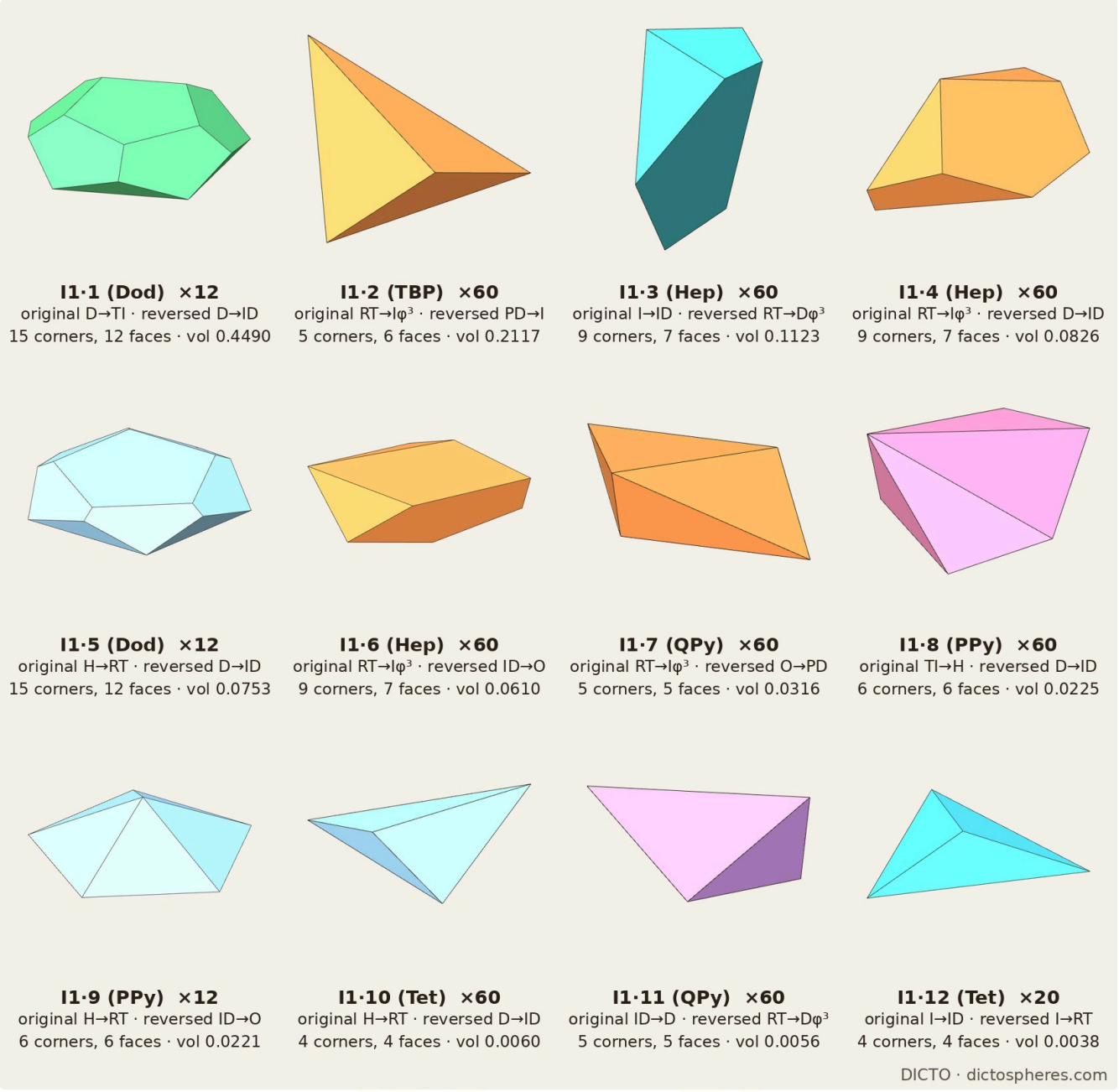


Figure 3. The 12 pieces of Interlock set 1 (ends setting), one piece of each, seen from the side with the outward direction up, each at its own scale, coloured by the original layer it lies in.

5. Interlock set 2 (Dharma-Middle): 502 pieces, 11 kinds

At $\rho^2 = 3\varphi^2/4$ the reversed cycle is exactly 3 times its ends-setting size, so its pieces are the ends pieces scaled by 3. Here the outer roofs pass through whole: the 30 roofs (RT → I·φ^3), the 20 truncated icosahedron → Hecatodihedron pyramids and the 20 icosidodecahedron → dodecahedron tetrahedra lie each inside one reversed piece. The cutting moves to the middle layers: the dodecahedron → truncated icosahedron columns are cut into 132 pieces and the Hecatodihedron → triacontahedron layer into 180.

Kind	Count	Original layer	Reversed layer	Corners	Faces (number × sides)	Volume (exact)
I2·1 Hep	30	RT → I·φ³	RT → D·φ³	8	2 × 3, 5 × 4	371/72 − 47/24√5
I2·2 Hen	12	D → TI	ID → O	11	5 × 3, 5 × 4, 1 × 5	625/10944 + 289/10944√5
I2·3 Hep	60	I → ID	D → ID	8	2 × 3, 5 × 4	-1585/576 + 3643/2880√5
I2·4 HPy	20	TI → H	I → RT	7	6 × 3, 1 × 6	-33/8 + 15/8√5
I2·5 Hep	60	D → TI	PD → I	8	2 × 3, 5 × 4	-101/456 + 12/95√5
I2·6 Hx	60	I → ID	RT → D·φ³	7	3 × 3, 2 × 4, 1 × 5	203/72 − 56/45√5
I2·7 Tet	20	ID → D	O → PD	4	4 × 3	1/192 + 1/192√5
I2·8 QPy	60	H → RT	PD → I	5	4 × 3, 1 × 4	-1/24 + 1/40√5
I2·9 Tet	60	H → RT	I → RT	4	4 × 3	-93/152 + 21/76√5
I2·10 Tet	60	D → TI	O → PD	4	4 × 3	1/576 + 1/576√5
I2·11 Tet	60	H → RT	I → RT	4	4 × 3	-23/304 + 11/304√5

Table 2. Interlock set 2, the middle setting, as Table 1.

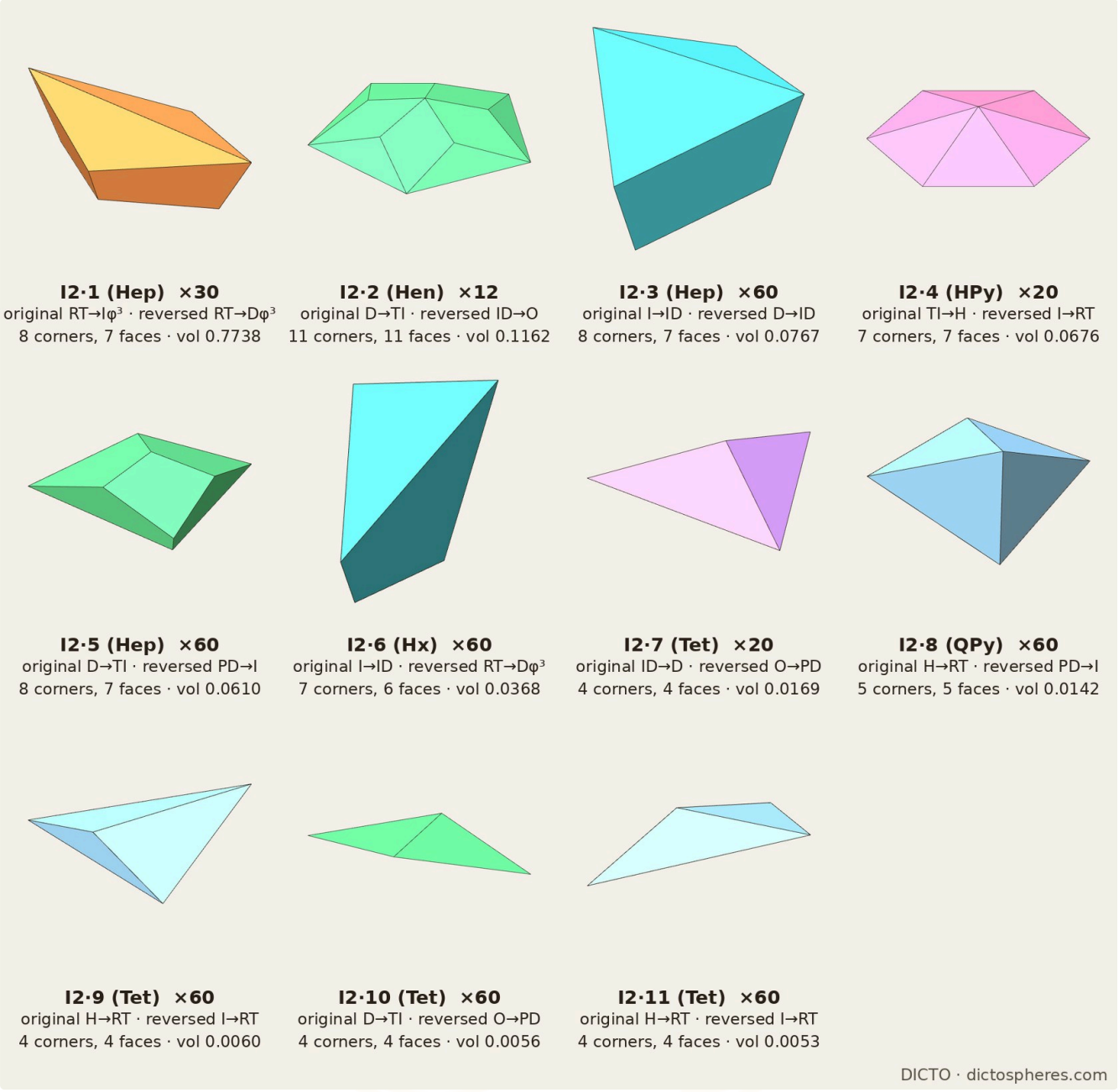


Figure 4. The 11 pieces of Interlock set 2 (middle setting), as Figure 3.

6. What the toggle moves

- **Where the cycles cut each other.** The ends setting cuts the outer roofs the most and leaves the middle columns whole; the middle setting leaves the roofs whole and cuts the middle layers.
- **One piece in both.** The 60 small Hecatodihedron → triacontahedron tetrahedra, $(-93/152 + 21\sqrt{5}/76)$ each, pass through whole in both settings (I1·10 = I2·9).
- **Equal volume, different shape.** I1·11 and I2·10 both have volume $(1 + \sqrt{5})/576$ but different shapes (5 and 4 corners).

7. Relation to known work, and status

Known and credited: polar reciprocation and the reversal of inclusion (convex geometry; Firsching 2015); the common refinement of two convex dissections by pairwise intersection, a standard construction; nested icosahedral shells in clusters and viruses (Takakura et al. 2007;

Twarock, Valiunas and Zappa 2015). **Not found** in our reference library and web searches (16 October 2026; printed catalogues not read): a cycle of nested solids and its polar reversal placed in one space; the two exact toggle settings and the proof that they exclude each other; the exact interlock dissections and their kinds.

8. Reproducibility

In studies/2026-10-16-interlock: `chains.py` (both chains in one frame), `interlock.py` (the pairwise intersections, float; arguments for the middle setting), `interlock_exact.py` (exact corners, sums and kinds → `interlock-ends-exact.txt`, `interlock-middle-exact.txt`), `interlock_kinds.py` (Tables 1 and 2 → `interlock-kinds.txt`), `interlock_kinds_render.py`, `render_kinds.html`, `shoot_kinds.mjs` and `kinds_gallery.py` (Figures 3 and 4). The toggle: `toggle_render.py`, `render_toggle.html`, `shoot_toggle.mjs` and `radial_strip.py` in studies/2026-10-16-polar-dharma. Checks: `python3 studies/_lib/test_geomcheck.py`.

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6. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
7. DICTO, Inverted Shells: The Spheres of Dharma Turned Inside Out, DICTO DOI: 10.5281/zenodo.23405029.